

ArrayList

What is it ?

- **Array** which can grow size
- Have length and capacity
- First index at 0, last index at length - 1

How does it work ?

- It solve insert problem of traditional array
- When we want to add new element, write the position length then update the length
- When we want to delete last element, reduce the length, also can clear that removed
- When we reach our capacity limit, create new array with larger capacity, (often doubled) then copies existing elements to the new array (BOOM! Dynamic grow 🙌
(^▽^)(👉))
- Good for stack-like operation, that's $O(1)$
- Insert or remove from beginning or middle is $O(n)$, require shift all elements

What is it good point ?

- Constant time accessing value by index, doing push/pop at the end $\llcorner(\rightarrow_{\wedge} \leftarrow)\gg$
- Slow at the beginning